ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems

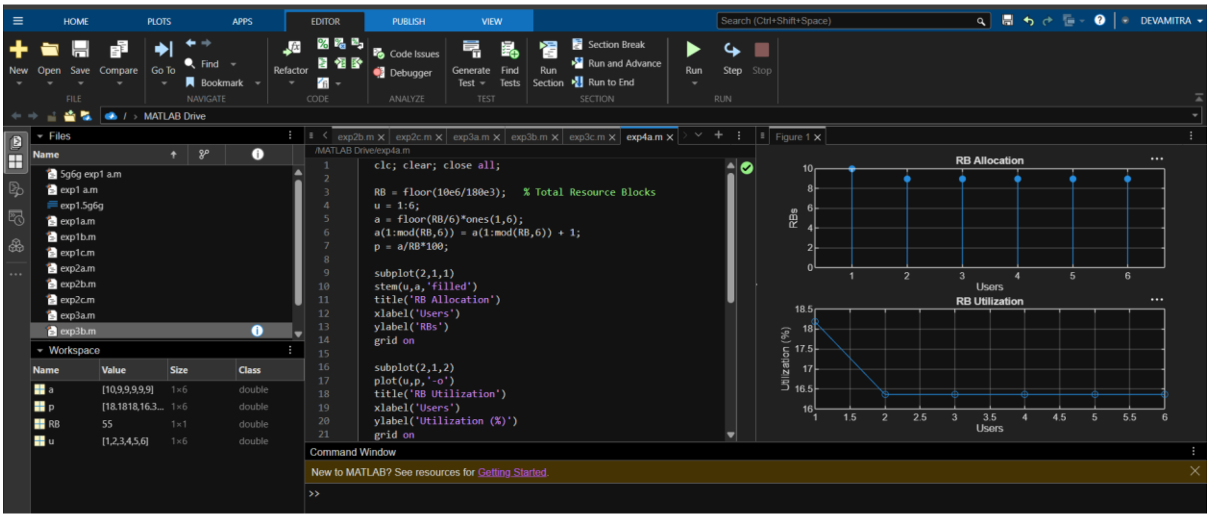
Ex.No.4: OFDMA Resource Block Allocation Date

Objective 1 Matlab Code and Output:

Program:

```
clc; clear; close all;
RB = floor(10e6/180e3);   % Total Resource Blocks
u = 1:6;
a = floor(RB/6)*ones(1,6);
a(1:mod(RB,6)) = a(1:mod(RB,6)) + 1;
p = a/RB*100;
subplot(2,1,1)
stem(u,a,'filled')
title('RB Allocation')
xlabel('Users')
ylabel('RBs')
grid on
subplot(2,1,2)
plot(u,p,'-o')
title('RB Utilization')
xlabel('Users')
ylabel('Utilization (%)')
grid on
```

Output:



Objective 2 Matlab Code and Output:

Program:

clc; clear; close all;

% Total system resource blocks

total_RB = 50;

% Number of active users

users = 1:5;

% RB allocation to users (example scenario)

alloc = [12 10 9 8 11];

% Resource Block Utilization (%)

util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')

disp(util')

figure

% -------- Graph 1: RB Allocation --------

subplot(2,1,1)

plot(users, alloc, '-o', 'LineWidth', 2)

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on

% -------- Graph 2: Utilization (%) --------

subplot(2,1,2)

plot(users, util, '-s', 'LineWidth', 2)

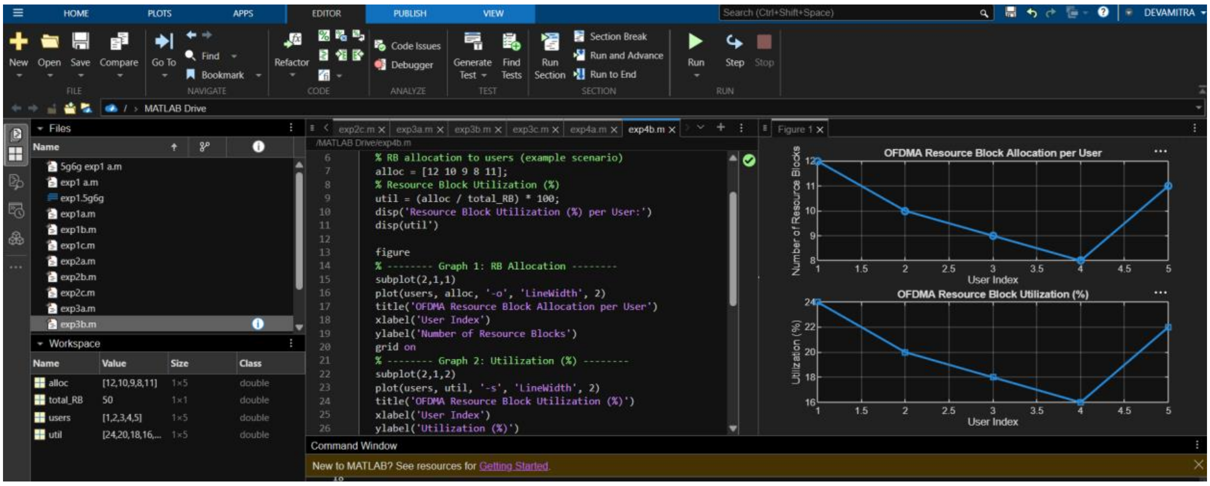
title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on

Output:



Objective 3 Matlab Code and Output:

Program:

clc; clear; close all;

% System parameters

rb_bw = 180e3; % Bandwidth per RB (180 kHz)

users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

```
% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')

disp(throughput')

figure

% -------- Graph 1: Throughput per User --------

subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2: RB vs Throughput --------

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

title('Resource Blocks vs Throughput Relationship')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on
```

Output: